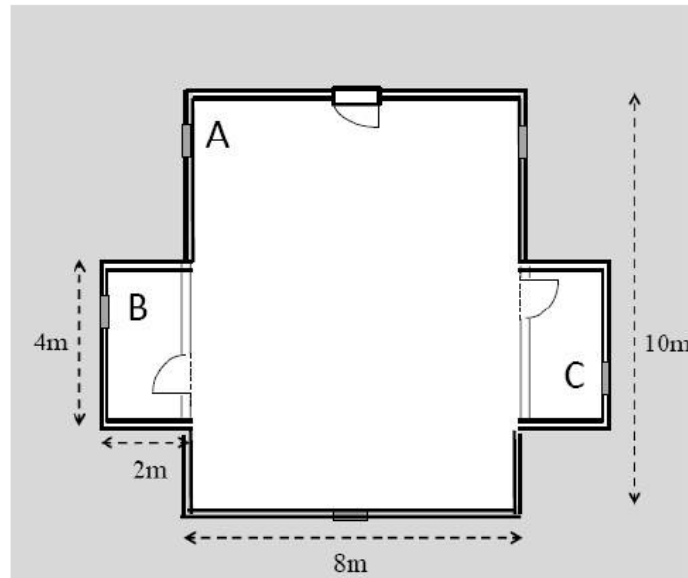


Temperature control of a three-rooms building

Consider the system illustrated in the following figure, consisting of a three-rooms building.



Model

$$\begin{aligned}c\rho V_A \dot{T}_A &= s_r u(T_B - T_A) + s_r u(T_C - T_A) + s_A U(T_E - T_A) + q_A \\c\rho V_{B,C} \dot{T}_B &= s_r u(T_A - T_B) + s_{B,C} U(T_E - T_B) + q_B \\c\rho V_{B,C} \dot{T}_C &= s_r u(T_A - T_C) + s_{B,C} U(T_E - T_C) + q_C\end{aligned}$$

c : specific heat of the air; V_i , $i = A, B, C$: Volume of room i ; s_r : wall surface between A and B (and C); s_A : wall surface between A and the environment; $s_{B,C}$: wall surface between B (and C) and the environment; u , U : transmittances; q_i , $i = A, B, C$: inputs (heating power).

- Equilibrium condition: $T_E = 0^\circ \text{ C}$, $T_A = T_B = T_C = \bar{T} = 20^\circ$,
 $q_A = \bar{q}_A = s_A \cdot U \bar{T} \text{ W}$, $q_B = q_C = \bar{q}_B = \bar{q}_C = s_{B,C} \cdot u \bar{T} \text{ W}$.
- We define: $\delta T_A = T_A - \bar{T}$, $\delta T_B = T_B - \bar{T}$, $\delta T_C = T_C - \bar{T}$, $\delta q_A = q_A - \bar{q}_A$,
 $\delta q_B = q_B - \bar{q}_B$, and $\delta q_C = q_C - \bar{q}_C$.

Around the given equilibrium condition:

$$\begin{bmatrix} \dot{\delta T}_B \\ \dot{\delta T}_A \\ \dot{\delta T}_C \end{bmatrix} = \begin{bmatrix} -(\Gamma + \gamma_r) & \Gamma & 0 \\ \gamma & -(2\gamma + \gamma_A) & \gamma \\ 0 & \Gamma & -(\Gamma + \gamma_r) \end{bmatrix} \begin{bmatrix} \delta T_B \\ \delta T_A \\ \delta T_C \end{bmatrix} + \begin{bmatrix} \frac{1}{c\rho V_{B,C}} & 0 & 0 \\ 0 & \frac{1}{c\rho V_A} & 0 \\ 0 & 0 & \frac{1}{c\rho V_{B,C}} \end{bmatrix} \begin{bmatrix} \delta q_B \\ \delta q_A \\ \delta q_C \end{bmatrix}$$

$$\gamma = \frac{s_r u}{c\rho V_A}, \Gamma = \frac{s_r u}{c\rho V_{B,C}}, \gamma_A = \frac{s_A U}{c\rho V_A}, \text{ and } \gamma_r = \frac{s_{B,C} U}{c\rho V_{B,C}}.$$

The system “centralized” model

$$\dot{x} = Ax + Bu$$

is given in the corresponding MATLAB file, where $x = [\delta T_A, \delta T_B, \delta T_C]^T$ and $u = [\delta q_A, \delta q_B, \delta q_C]$. We assume that all the state variables are measurable.

Problem:

1. Decompose the state and input vectors into subvectors, consistently with the physical description of the system. Obtain the corresponding decomposed model.
2. Generate the system matrices (both continuous-time and discrete-time, the latter with a sampling time selected compatibly with the continuous-time dynamics). Perform the following analysis:
 - a. Compute the eigenvalues and the spectral abscissa of the (continuous-time) system. Is it open-loop asymptotically stable?

- b. Compute the eigenvalues and the spectral radius of the (discrete-time) system. Is it open-loop asymptotically stable?
- 3. For different state-feedback control structures (i.e., centralized, decentralized, and different distributed schemes) perform the following actions
 - a. Compute the continuous-time fixed modes (if necessary, adjust the rounding tolerance in the dedicated function)
 - b. Compute the discrete-time fixed modes (if necessary, adjust the rounding tolerance in the dedicated function)
 - c. Compute, if possible, the CONTINUOUS-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
 - d. Compute, if possible, the DISCRETE-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
 - e. Analyze the properties of the so-obtained closed-loop systems (e.g., stability, eigenvalues) and compute the closed-loop system trajectories (generated both in continuous-time and in discrete-time) of the room temperatures starting from a common random initial condition.